

ABM & Urban Heat

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Think about the feeling of walking around in...

This ↓



Versus this ↓



What makes urban heat complex?



Research Question

How do different spatial distributions and quantities of various land covers impact heat interactions in an urban environment?



Hypothesis

Areas with more grass that are better distributed will stay cooler, while areas dominated by concrete or poorly arranged land covers will trap more heat.

Model Components

Included Components

- Trees
- Grass
- Concrete

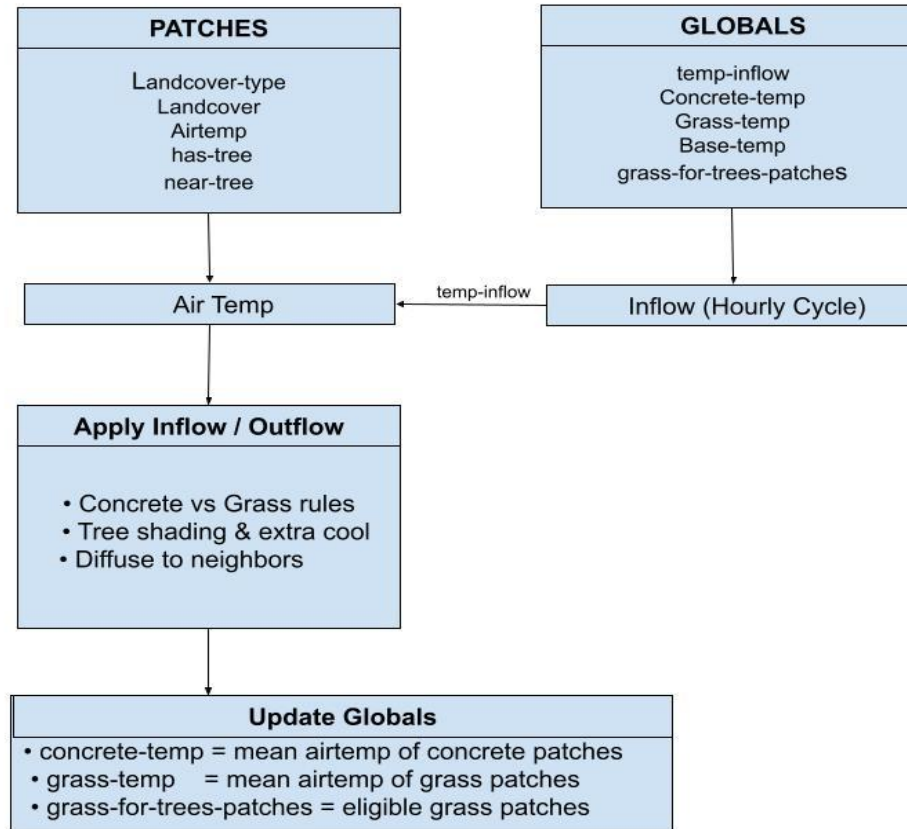
Why?

Excluded Components

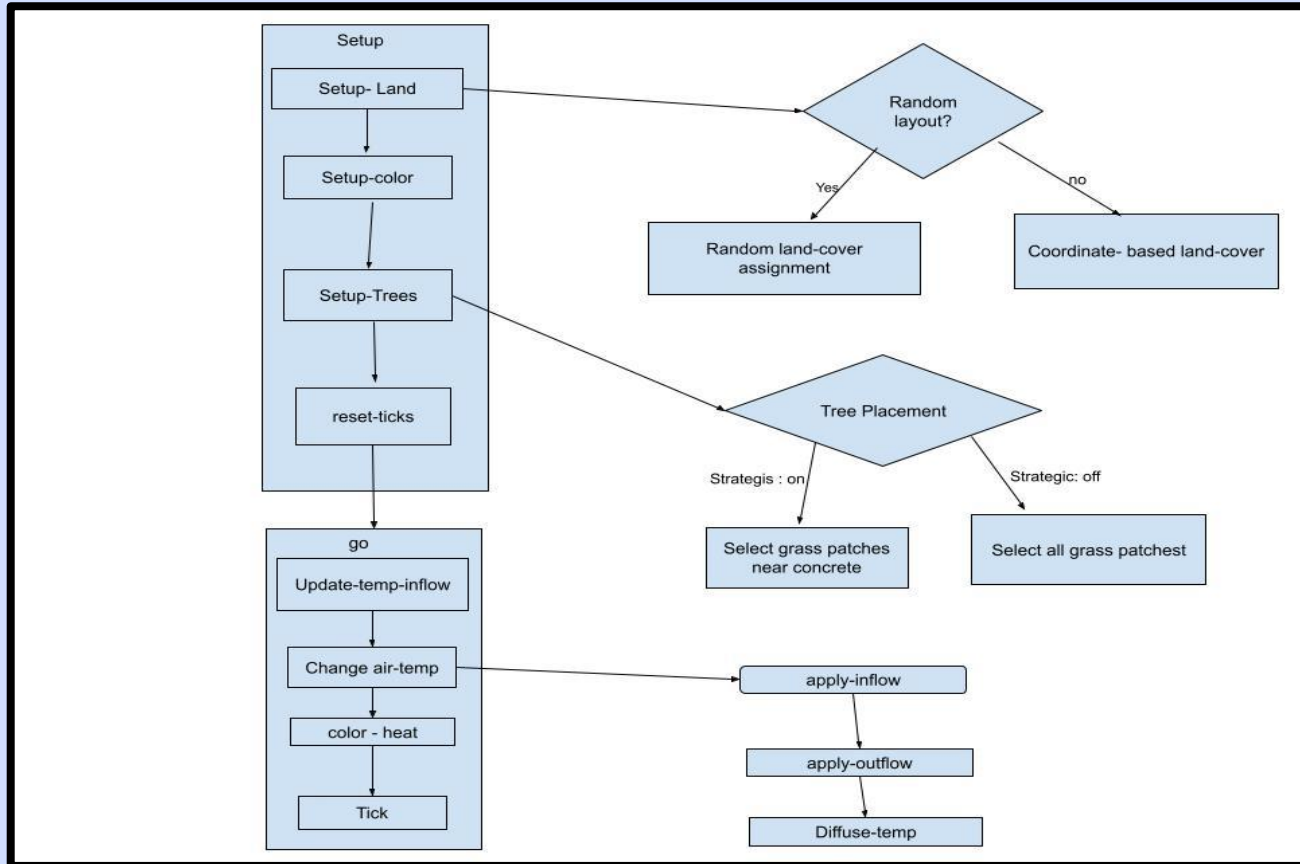
- Appliances
- Traffic
- Buildings

Conceptual Diagram

○



Model Order of Events Diagram



Parameterization

Default Parameters	
inflow-rate	1.0
%-inflow-removed	0.05
outflow-grass	0.50
outflow-concrete	0.43
tree-outflow	0.20
diffuse-slider	0.10



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The effect of tree shade and grass on surface and globe temperatures in an urban area

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Surface cover
Trees
Grass

ABSTRACT
The process of urbanisation alters the thermal balance of an area resulting in an urban heat island effect where cities can be several degrees centigrade warmer than the surrounding rural landscape. This increased heat can make cities uncomfortable places and, during heat waves, can pose serious health risks. This study looked at the role that trees and grass can play in reducing regional and local temperatures in urban areas during the summer within the urban landscape of Manchester, UK. In June and July 2009 and 2010, we monitored the surface temperatures of small plots composed of concrete and grass in the presence or absence of tree shading, and measured globe temperatures above each of the surfaces. The same measures were also recorded at mid-day on larger expanses of asphalt and grass in an urban park. Both surface and shade greatly affected surface temperatures. Grass reduced maximum surface temperatures by up to 24 °C, similar to model predictions, while tree shade reduced them by up to 19 °C. In contrast, surface composition had little effect upon globe temperatures, whereas shading reduced them by up to 5–7 °C. These results show that both grass and trees can effectively cool surfaces and so can provide regional cooling, helping reduce the urban heat island in hot weather. In contrast grass has little effect upon local air or globe temperatures, so should have little effect on human comfort, whereas tree shade can provide effective local cooling.

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Model Parameters

Scenario Parameters

Concrete

Clustering

tree-radius

num-trees

Generalizable Parameters

random-layout?

strategic-trees

Minimum

Maximum

Decline

Calibration Parameters

inflow-rate

%-inflow-removed

tree-outflow

grass-outflow

concrete-outflow

diffuse-slider

Scenario Parameters

Concrete 50

Clustering 10

num-trees 10

random-layout? ☐ On ☒ Off

strategic-trees ☐ On ☒ Off

Inflow Rate Sliders

inflow-rate 1.0

%-inflow-removed 0.85

Outflow Rate Sliders

outflow-grass 0.50

outflow-concrete 0.43

tree-outflow 0.20

Radius of patches
affected by trees

Diffusion Slider

tree-radius 4

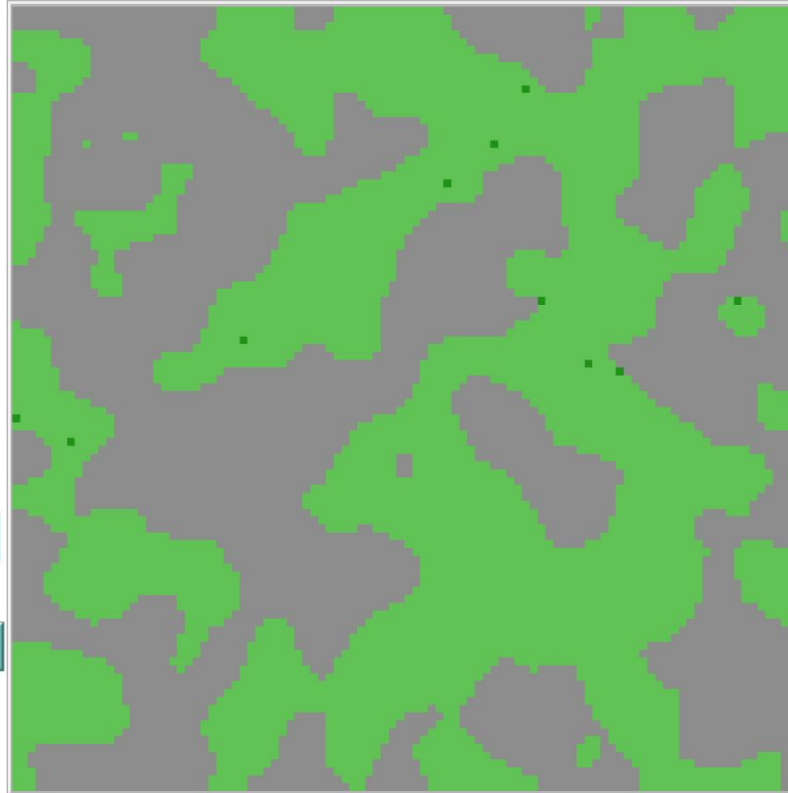
diffuse-slider 0.10

Temperature Sliders

minimum 68

maximum 80

decline 78



Overall mean Airtemp

0

setup

Overall mean Grass Te...

0

go

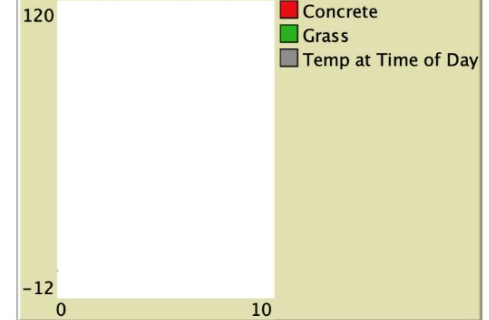
Overall mean Concrete ...

0

temp-inflow

0

Mean Airtemp over Ti...



Scenario Parameters

Concrete 50 ☐ On random-layout? ☐ Off

Clustering 10 ☐ On strategic-trees ☐ Off

num-trees 10

Inflow Rate Sliders

inflow-rate 1.0

%-inflow-removed 0.05

Outflow Rate Sliders

outflow-grass 0.50

outflow-concrete 0.43

tree-outflow 0.20

Radius of patches affected by trees

tree-radius 4

Diffusion Slider

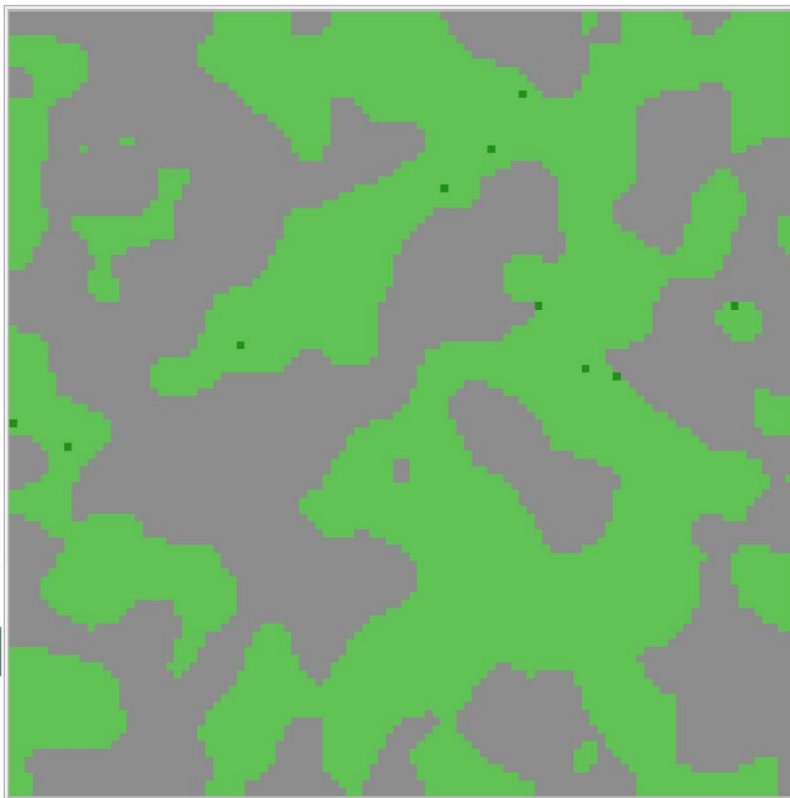
diffuse-slider 0.10

Temperature Sliders

minimum 68

maximum 80

decline 78



Overall mean Airtemp

0

setup

Overall mean Grass Te...

0

go

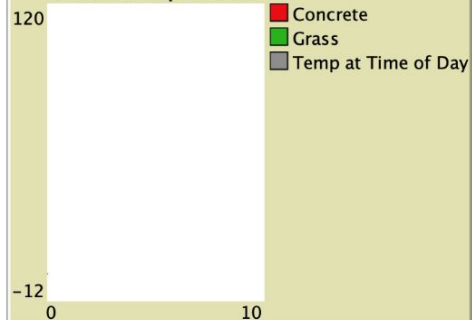
Overall mean Concrete ...

0

temp-inflow

0

Mean Airtemp over Ti...

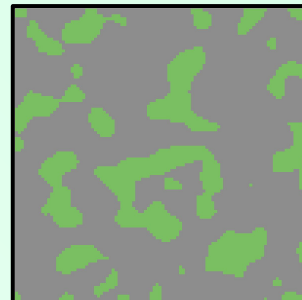
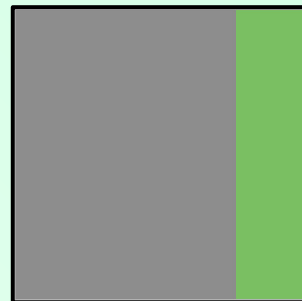
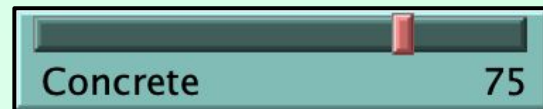
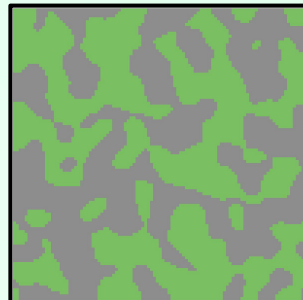
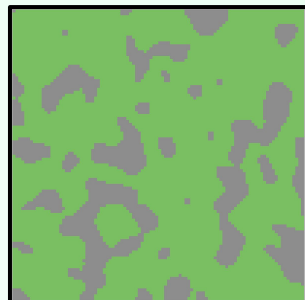
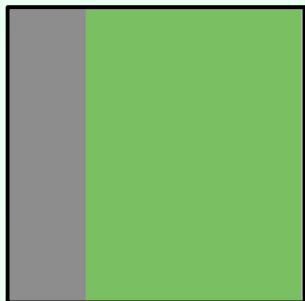


Background for Scenarios

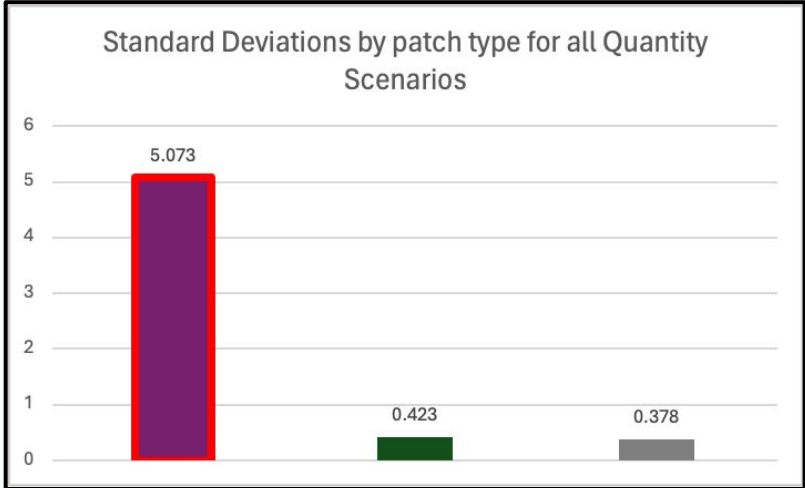
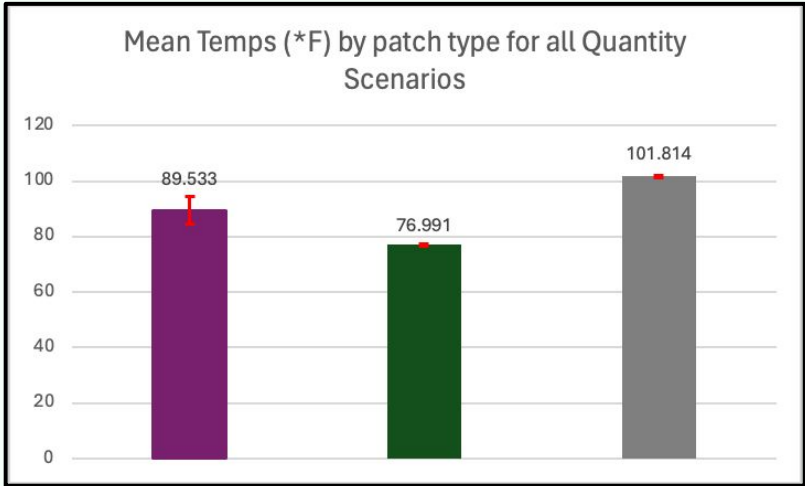
- All temperatures are in degrees Fahrenheit (sorry!)
- We are looking at temperatures from a day in Boston, but are only using the temperatures from peak time (2pm/14:00) for our analysis

While we agree that looking at Karachi was nice, it more demonstrated the generalizability of our model, rather than truly exploring the parameters and our research question more directly, so we will omit today.

Concrete / Quantity



*Clustering is 10



Patch Type Key

Overall
Grass
Concrete

Changing Parameters			
Concrete	Clusterin g	Number Trees	Radius
25/50/75	10	10	4

Scenario Results

Boston	Overall Mean (30 runs each)	Grass Mean (30 runs each)	Concrete Mean (30 runs each)
25% Concrete	83.335	77.31	101.408
50% Concrete	89.503	77.145	101.86
75% Concrete	95.761	76.517	102.174

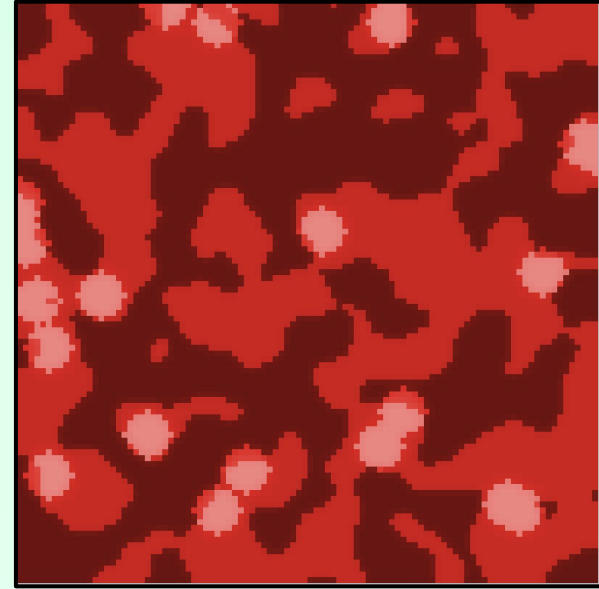
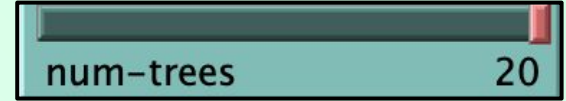
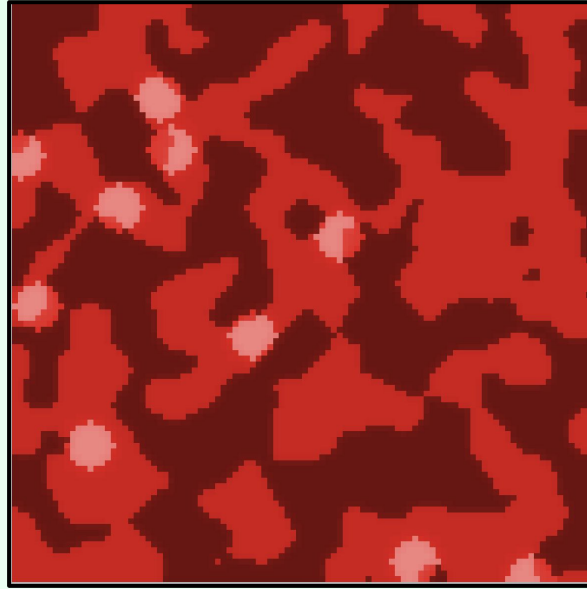
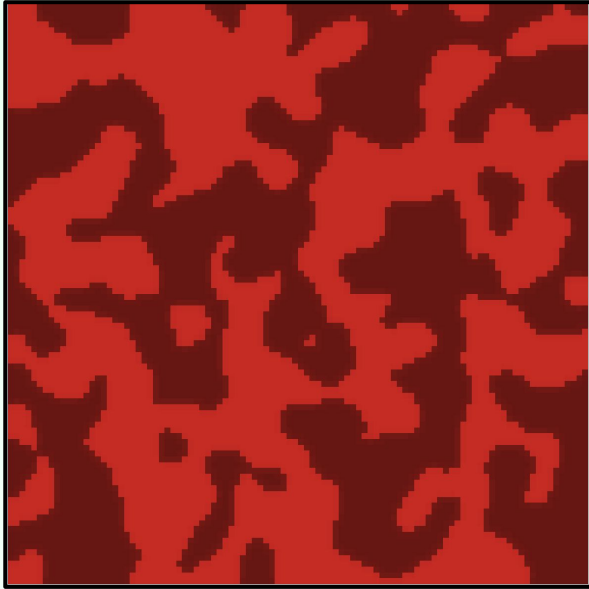
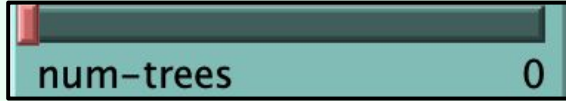
+12.426*F

-0.793*F

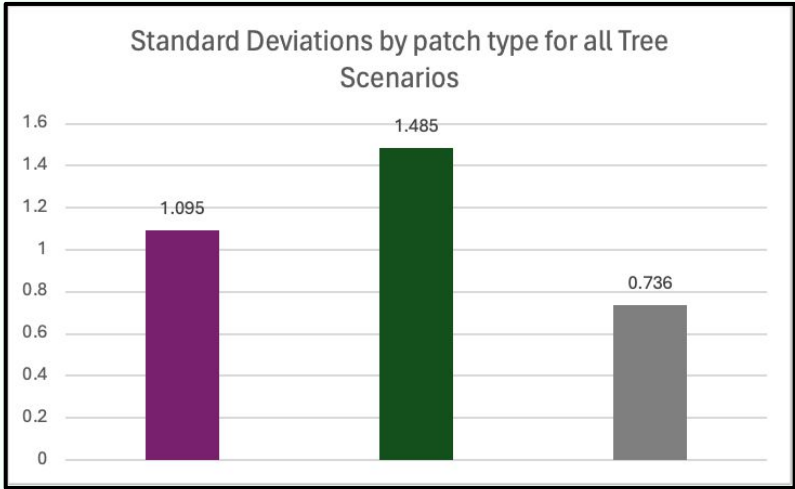
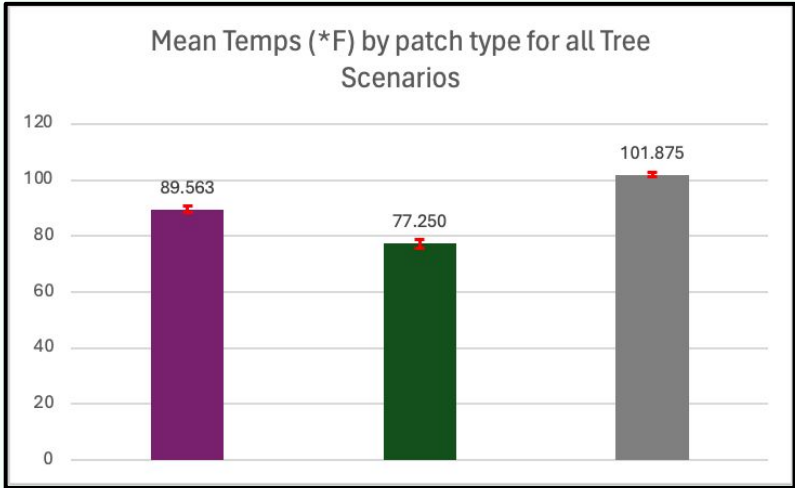
+0.766*F

In line with previous research, these results show how quantity of different land covers can be a driver of overall temperature in a given urban area.

Trees



*Clustering is 10



Patch Type Key

Overall
Grass
Concrete

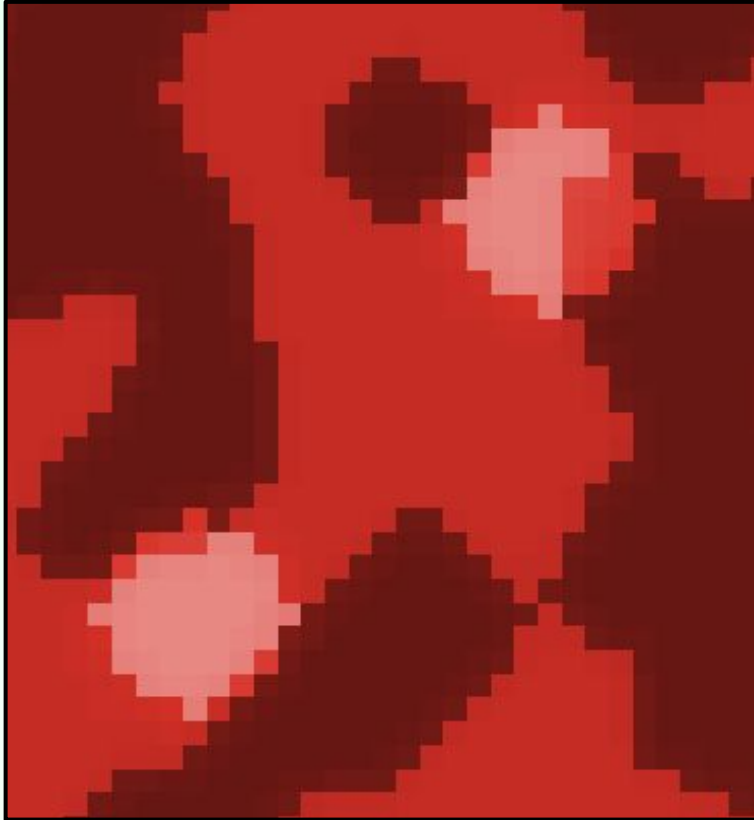
Changing Parameters			
Concrete	Clustering	Number Trees	Radius
50	10	0/10/20	4

Scenario Results

Boston	Overall Mean (30 Runs each)	Grass Mean (30 runs each)	Concrete Mean (30 runs each)
0 Trees	90.928	79.108	102.745
10 Trees	89.503	77.145	101.86
20 Trees	88.258	75.498	101.019
	-2.67°F	-3.61°F	-1.726°F

Overall, the model cooled when there are more trees, which is expected. Grass patches were more cooled by trees than concrete. Why?

It's how our trees are coded...



Radius of shading was set to 4 for our testing, but...

```
ask patches [  
  set near-tree any? patches in-radius 5 with [has-tree]  
]
```

Trees could be placed anywhere from 1-5 patches away from concrete, which aligns with the results that show a more pronounced cooling effect on grass patches.

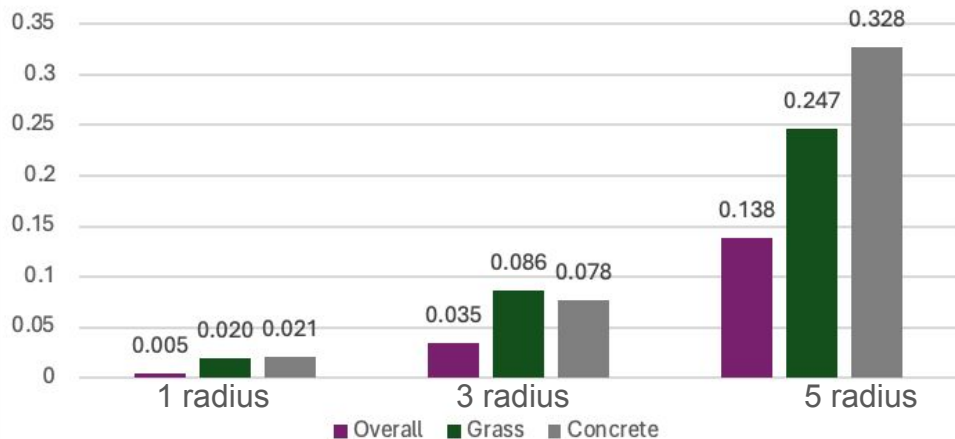
We should have caught this sooner and changed it, but could be place for further exploration.

Tree radius testing

Changing Parameters

Concrete	Clustering	Number Trees	Radius
50	10	10	1 / 3 / 5

Standard Deviations of patch types with different tree radius



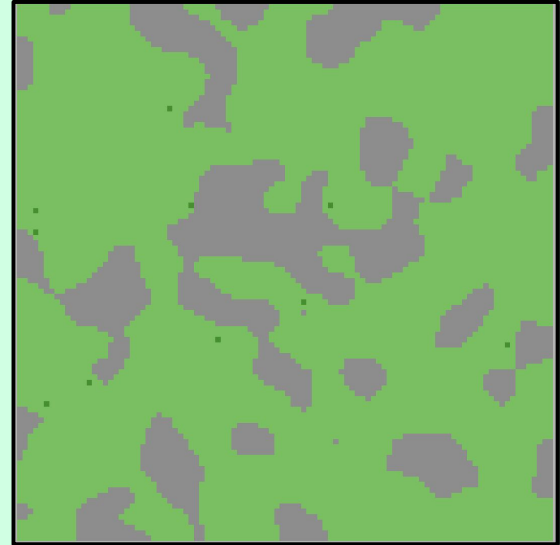
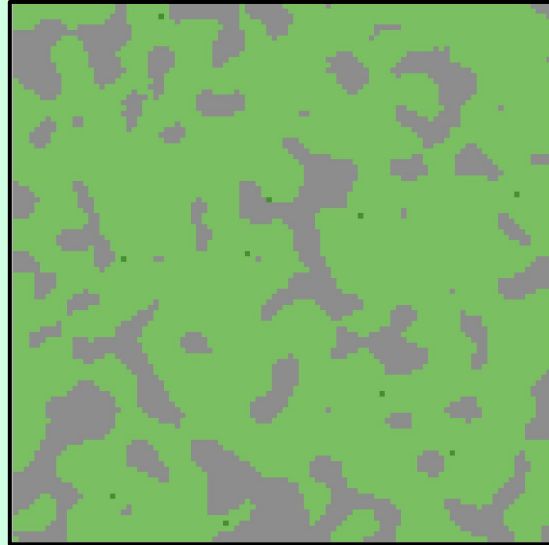
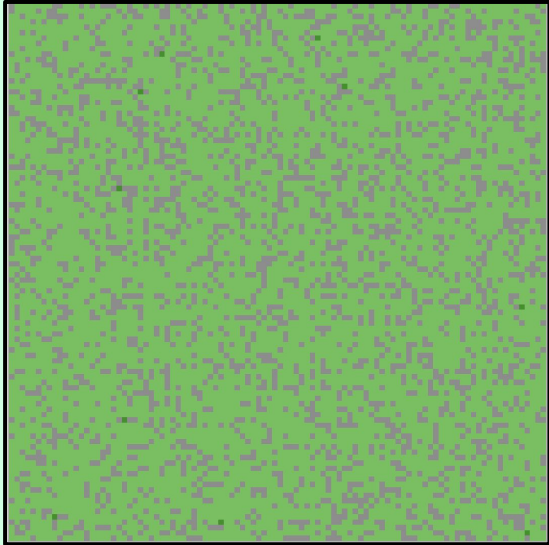
Boston	Overall Mean (30 Runs each)	Grass Mean (30 runs each)	Concrete Mean (30 runs each)
1 Radius	90.793	78.67	102.917
3 Radius	90.118	77.59	102.64
5 Radius	88.74	75.896	101.58

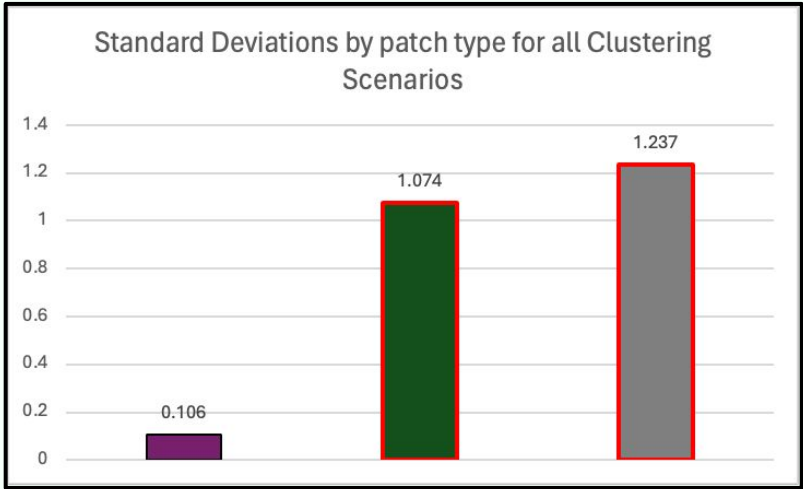
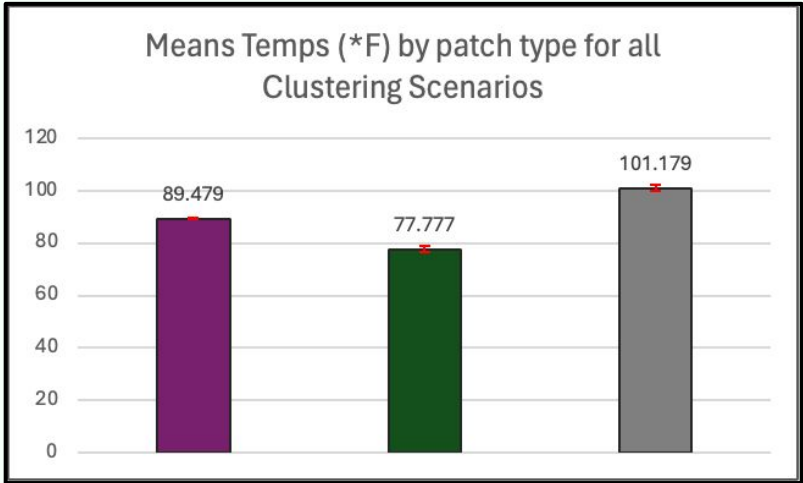
-2.05°F

-2.77°F

-1.34°F

Clustering





Patch Type Key

Overall
Grass
Concrete

Changing Parameters			
Concrete	Clustering	Number Trees	Radius
50	0/5/10	10	4

Scenario Results

Boston	Overall Mean (30 runs each)	Grass Mean (30 runs each)	Concrete Mean (30 runs each)
0 Clustering	89.365	<u>79.273</u>	<u>99.456</u>
5 Clustering	89.503	<u>77.145</u>	<u>101.86</u>
10 Clustering	89.569	<u>76.913</u>	<u>102.22</u>

+0.204*F

-2.36*F

+2.764*F

Grass and Concrete have noticeable differences in temperatures depending on the clustering level of 0, 5, or 10. **Can we explore this further?**

Range of Temperatures

Degrees added to difference between Grass & Concrete means	Boston	Ranges	Grass Mean (30 runs)	Concrete Mean (30 runs)
	0 Clustering	20.183	<u>79.273</u>	<u>99.456</u>
+ <u>4.532°F</u>	5 Clustering	24.715	<u>77.145</u>	<u>101.86</u>
+ 0.592 *F	10 Clustering	25.307	<u>76.913</u>	<u>102.22</u>

Range of differences between Grass & Concrete for different clustering scenarios

20.183

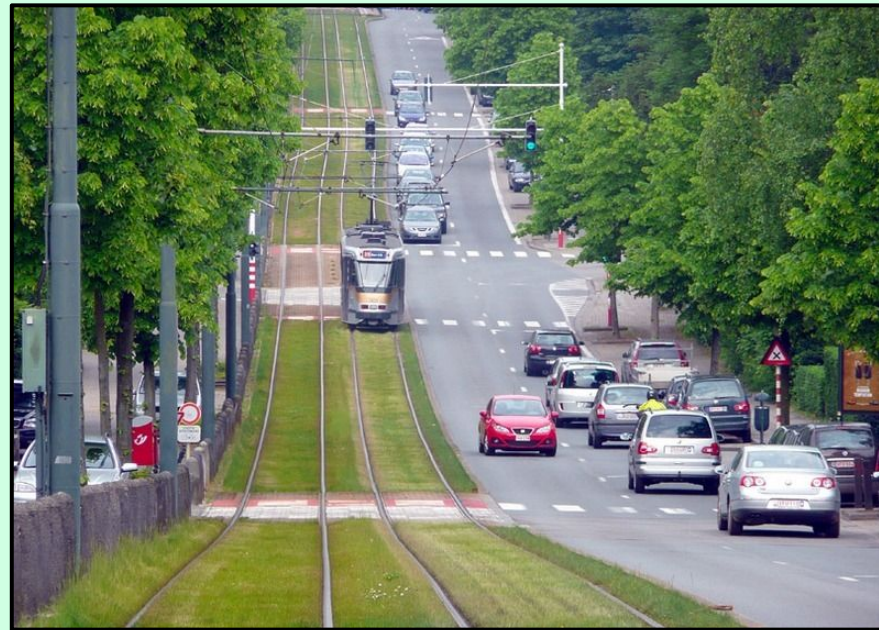
24.715

25.307

It seems that the jump from 0 to 5 clustering is what had the greatest effect on the means of grass and concrete patches. There was an increase at 10 clustering, but much less drastic for the same magnitude increase in clustering as from 0-5.

This leads us to a pretty interesting conclusion in regards to our research question....

Compare these...



Stormwater management & urban heat?



Limitations

- Trees take a long time to grow to a size that provides plentiful shade, which is why they are not one of our primary recommendations.
- The model does not take into account other drivers of urban heat, such as traffic, appliances, buildings, etc.
- We have no way of exactly matching an urban area with layout

Future Work

- Integrating people in some way, using health data, to see how people respond to different temperatures
 - Social disparities regarding to heat
- Generalizing to other cities
- Changing trees to be more variable in size and traits

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